FOR QUESTIONS

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AND RUST?

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Both support fast native code · both support compiled, generic code and vector operations

Safer by construction · safe Rust prevents some memory errors. Our C++ checks catch some errors at build time or during tests.

Our integration choice · C++ fits our Qt design. A Rust bridge would not always require copying data.

Rust would also be a reasonable option

WHY A PYTHON DUAL CLASS IS NOT THE SAME

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```cpp
template <typename T>
T gaussian(T A, T x, T x0, T sigma) {
    T d = x - x0;
    return A * exp(-d * d / (T(2) * sigma * sigma));
}
```

PYTHON `class Dual`

each ∗ is a method lookup,

a new heap object, reference counting

≈ 200 – 500 ns per operation

C++ `template<T>`

a dual is two doubles in CPU registers,

the compiler can inline the operators

≈ 1 ns per operation

JAX also supports forward-mode derivatives. C++ is one implementation choice.

AUTOMATIC DIFFERENTIATION

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A number becomes a pair: (value, slope)

$y = x^2$ at $x = 3$: $(3, 1) \times (3, 1) = (9, 6)$

AUTOMATIC DIFFERENTIATION

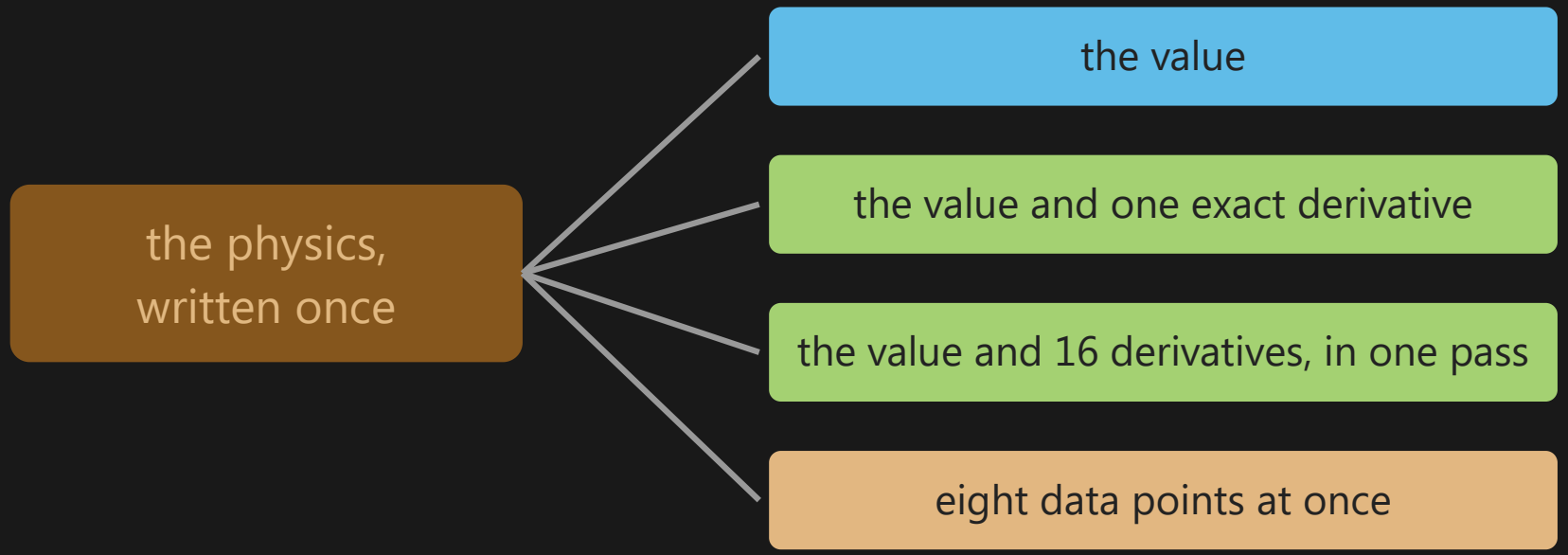
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The calculus lives once, in about 15 primitive rules · the physics never sees it

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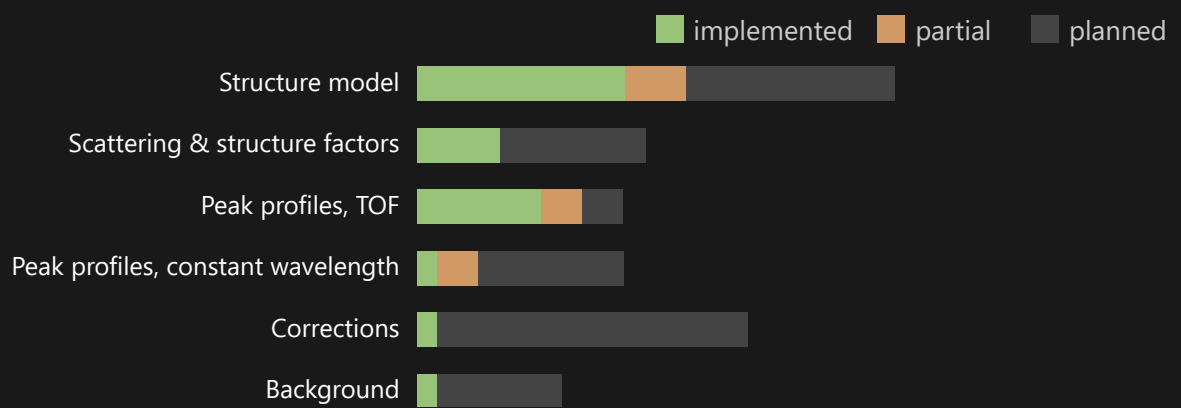


No separate derivative formulas · the compiler builds all four from the same formula

HOW COMPLETE IS THE ENGINE?

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implemented / partial / planned, by area



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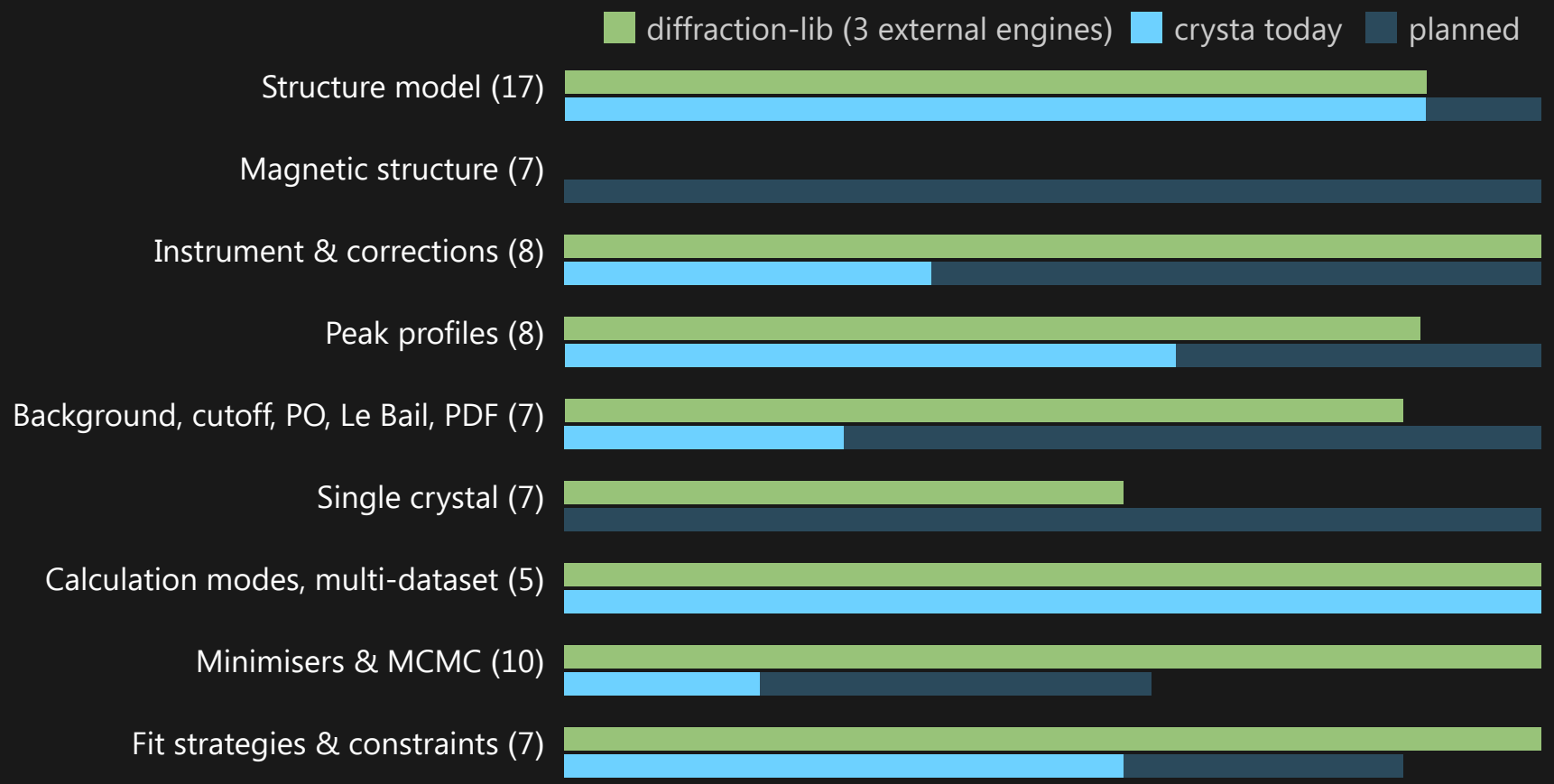
implemented / partial / planned, by area

Today: the TOF neutron powder path, end to end, verified against FullProf

Constant wavelength on the public API · everything else documented and planned

CRYSTA VS EASYDIFFRACTION TODAY

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Engine-level features only · diffraction-lib features page vs crysta feature catalog, 2026-09-03 · bar = share of the group

THE FULL STACK

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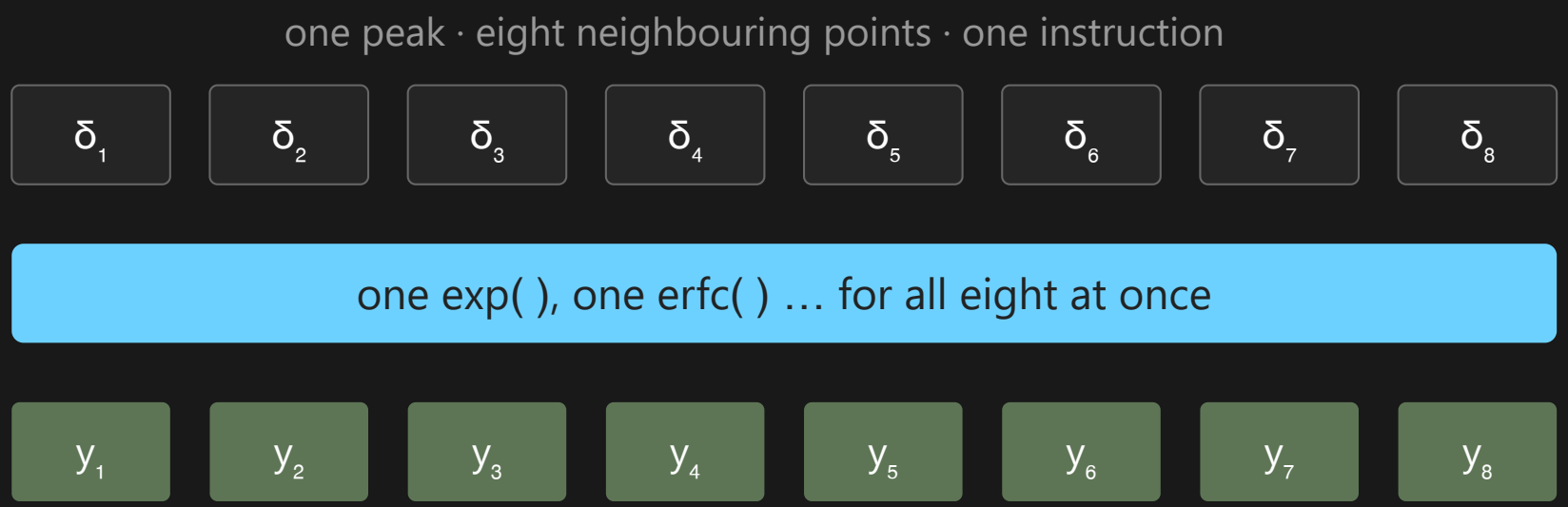
C++20 | nanobind | Qt / QML | Qt for WebAssembly | Eigen | SLEEF | OpenMP | CMake | pixi

SIMD widths: AVX-512 (8 lanes), AVX2 (4), NEON (2), chosen at run time · SLEEF for vector exp, erfc, sin, cos, log, atan2

Reproducibility gates in CI: SIMD width 0 / 4 / 8, threads 1 to 64, link-time optimisation on or off · byte-identical output

EVERY CORE, EIGHT NUMBERS AT A TIME

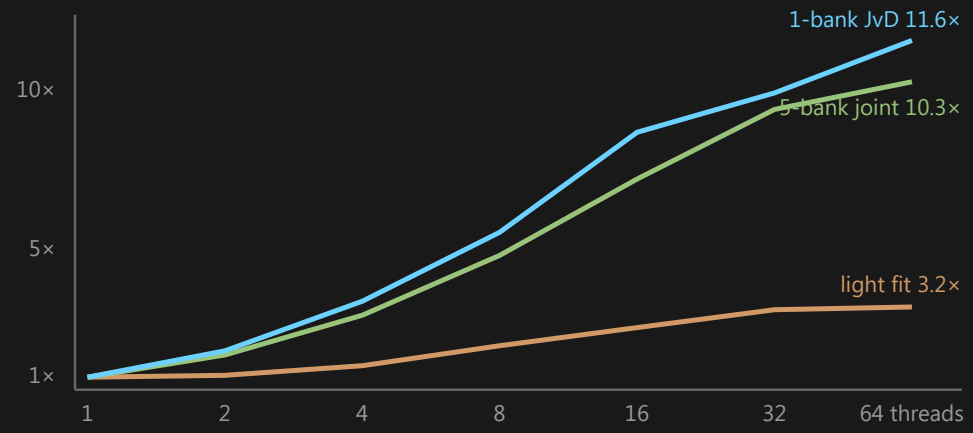
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SIMD · peak kernel 3.1× · whole fit 1.5×

EVERY CORE, EIGHT NUMBERS AT A TIME

speed-up vs threads · 64-core Intel Xeon



Threads · one fit on 64 cores, up to 11× · and the answer is identical at 1 or 64

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1 or 64 threads · 2, 4 or 8 numbers per instruction · tested compiler settings

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Identical numbers, to the last digit,
in the tested configurations

checked automatically on every change · these are the numbers that go into your paper

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Graphics cards · Numba, JAX and PyTorch have a GPU path today

Exploration · a new profile function in an afternoon

Orchestration · a thousand datasets, notebooks, plots

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The engine is C++ · the workflow around it is Python

MEASURE THE PROCESS, DELETE THE RITUAL

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median time to accept · 73 finished task cycles



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The planning stage lost its default · repeated plan changes can signal unclear requirements

Every number an agent reports names its source · no provenance, no number

NOT THERE YET

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Nothing technically stops an agent from merging · discipline, not a lock

No daily health check across the repositories

No AI inside the product yet · assistants and what-if refits are drafted, not built

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And the cost is not tracked · we measure hours, not tokens

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AI agents wrote the new code under a process with human oversight.

WHAT IT LOOKS LIKE

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SCREENSHOT PLACEHOLDER

The web app or desktop app on a real dataset: pattern, residual, parameter table.
The browser version still needs to be built and tested.

RISKS AND CONCERNS

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Incorrect or incomplete requirements propagate into the implementation

Different LLMs may share the same wrong assumptions

Overcomplicated or hard-to-maintain solutions

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Overcomplicated or hard-to-maintain solutions

LLM and provider availability, price changes

Human review becomes the bottleneck if every generated line must be read

OPPORTUNITIES

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Higher development productivity

Developer effort moves from implementation details to specifications, architecture, scientific validation and user experience

Faster exploration and prototyping of alternative ideas

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A separate AI review may find problems before human review

Larger projects become feasible with limited developer resources

SMALL, INSTANT, EVERYWHERE

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1.2 MB

native binary, without Python

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The design shares one engine across four interfaces. The browser build is planned.

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An unjustified weaker test is a blocking review finding

a loosened tolerance, a deleted assertion, an added skip

WRITTEN DOWN, OR IT DID NOT HAPPEN

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```
r12  implement:  green   — all hidden tests pass
r13  review:     reject  — a tolerance was loosened
r15  review:     accept
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A verdict ledger: one committed row per completed turn, append-only

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A missing or invalid record makes this check stop

I DELIBERATELY DID NOT READ THE CODE

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// x>= 25: the direct product overflows (erfc underflows), so the asymptotic form
inline double erfcx_nonneg(double x) {
    if (x < 25.0) {
        return std::exp(x * x) * std::erfc(x);
    }
    const double t = 1.0 / (x * x);
    return (std::numbers::inv_sqrtpi / x) *
           (1.0 + t * (-0.5 + t * (0.75 + t * (-1.875 + t * 6.5625))));
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My work moved from writing it to specifying it and proving it wrong

BY THE NUMBERS

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10	151	44
weeks	tasks shipped	milestones

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weeks	tasks shipped	milestones

202	505	≥ 80 %
review rejects	surprises logged	test coverage

Source: our roadmap and measurement records, 3 September 2026 · first commit 25 June 2026

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① Write the specification myself, then alternate: AI refines, I correct, AI plans, I correct, AI implements, I review and test

works best · costs by far the most of my time

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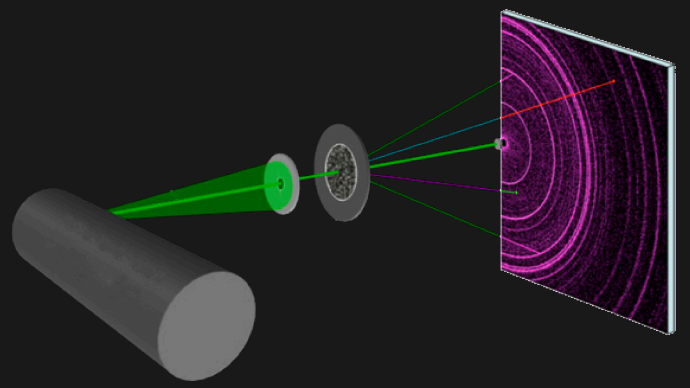
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③ Now · looking for the balance: a correct result with less manual work

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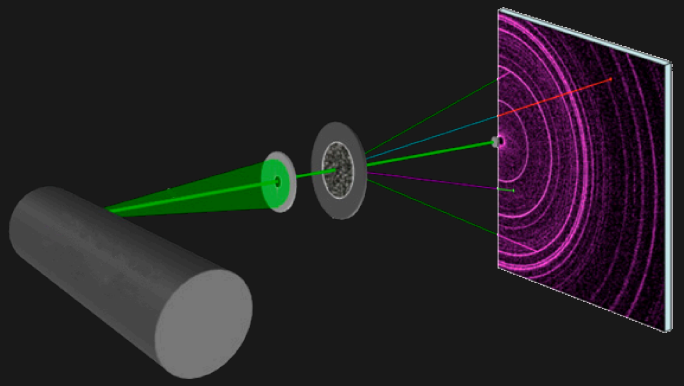


Measure

a beam, a powder, a detector

NICOS

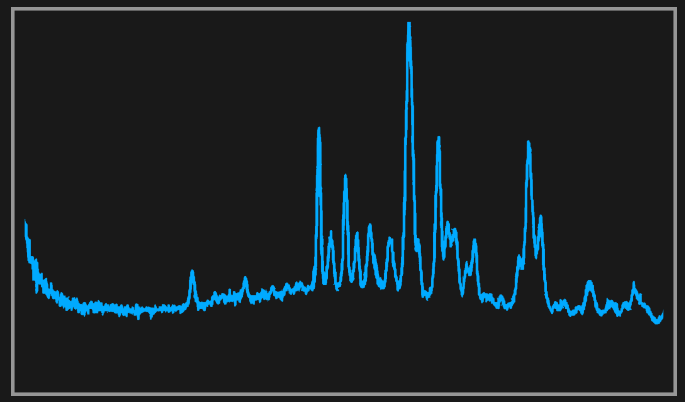
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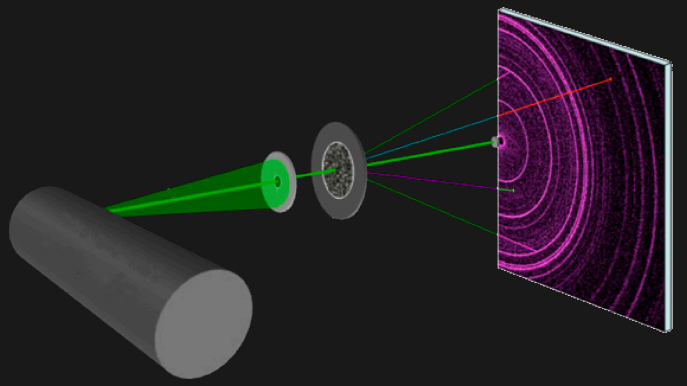


Reduce

the rings become one curve

scipp

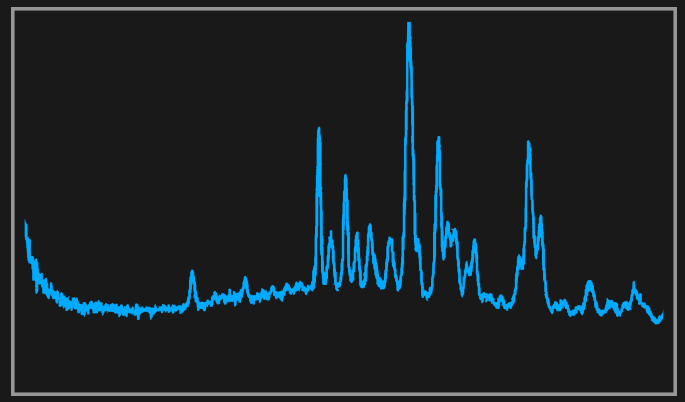
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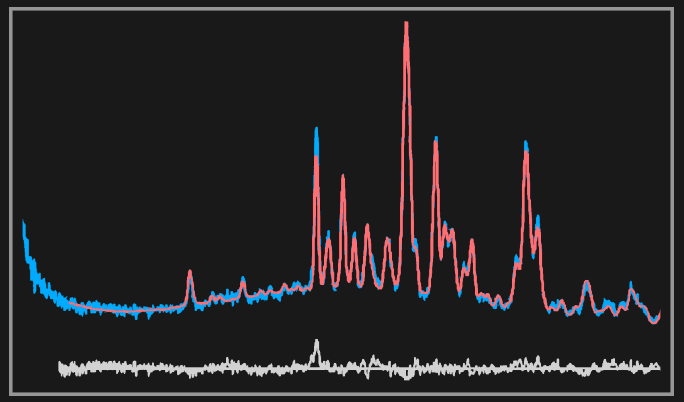
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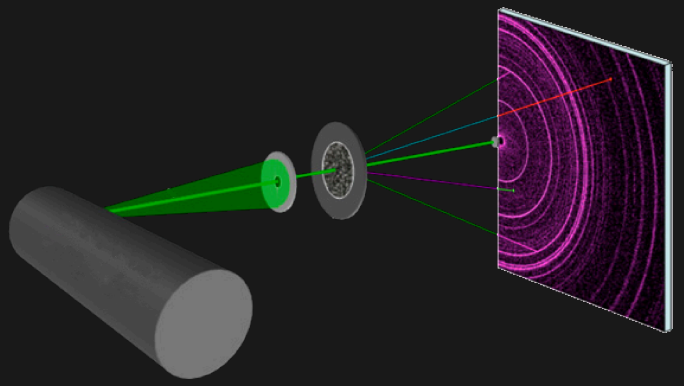


Refine

compute it from a model and adjust

easy diffraction

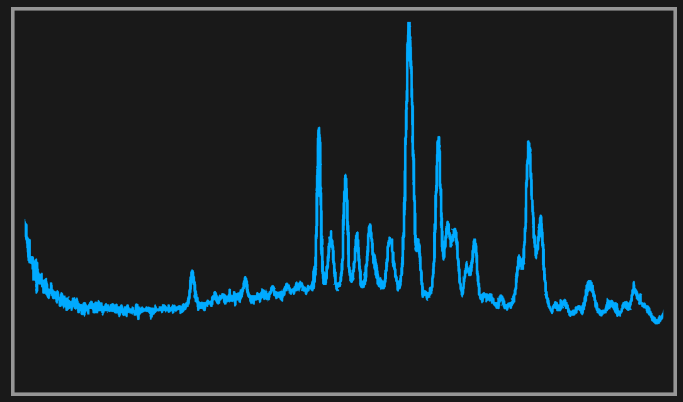
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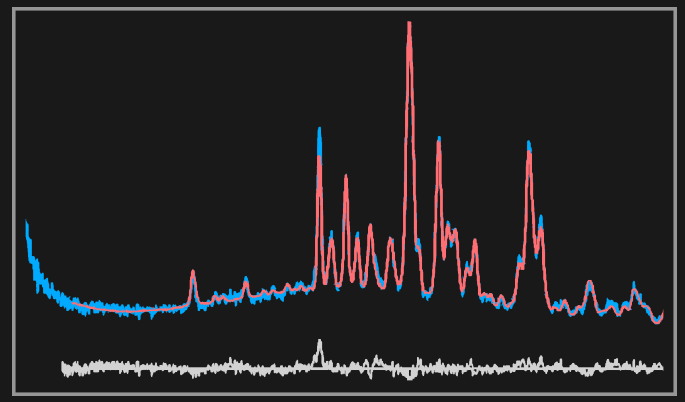
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sciPP



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That third step runs again and again — and that is where the time goes

WHO IS IT FOR

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- Python library · scripting, Jupyter, teaching material
  - users who work with scripts

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Command line · no Python, no GUI

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Web app · the desktop app in a browser

intended for quick checks and everyday work

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came from code AI agents wrote
and that I have never read